Markov Chains (AI HL)

ClipSAT Printable Practice Suite (clipsat.org) • Topic 6.03

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Transition Matrices and State Vectors

Are $\mathbf{u} = (2; 0; 0)$ and $\mathbf{w} = (7; 0; 0)$ parallel? Show your complete algebraic work.

Final Answer: Yes. Working through this vectors step: Check if $\mathbf{w} = k\mathbf{u}$ for a scalar k: $k = 3.5$ works, so they are parallel.

2 [Foundational] • Target: Multi-Step Transitions

Are $\mathbf{u} = (1; -1; 3)$ and $\mathbf{w} = (-2; 2; -5)$ parallel? Show your complete algebraic work.

Final Answer: No. Working through this vectors step: Check if $\mathbf{w} = k\mathbf{u}$ for a scalar k: this is false, so they are not parallel.

3 [Foundational] • Target: Steady-State (Stationary) Distribution

Are $\mathbf{u} = (2; -4; 1)$ and $\mathbf{w} = (-4; 8; -1)$ parallel? Show your complete algebraic work.

Final Answer: No. Working through this vectors step: Check if $\mathbf{w} = k\mathbf{u}$ for a scalar k: this is false, so they are not parallel.

4 [Foundational] • Target: Absorbing Markov Chains

Are $\mathbf{u} = (-2; -4; 0)$ and $\mathbf{w} = (6; 13; 0)$ parallel? Show your complete algebraic work.

Final Answer: No. Working through this vectors step: Check if $\mathbf{w} = k\mathbf{u}$ for a scalar k: this is false, so they are not parallel.

5 [Foundational] • Target: Transition Matrices and State Vectors

Are $\mathbf{u} = (2; 4; 0)$ and $\mathbf{w} = (-6; -12; 0)$ parallel? Show your complete algebraic work.

Final Answer: Yes. Working through this vectors step: Check if $\mathbf{w} = k\mathbf{u}$ for a scalar k: this is true, so they are parallel.

6 [Foundational] • Target: Multi-Step Transitions

Are $\mathbf{u} = (0; -4; 0)$ and $\mathbf{w} = (0; 12; 0)$ parallel? Show your complete algebraic work.

Final Answer: Yes. Working through this vectors step: Check if $\mathbf{w} = k\mathbf{u}$ for a scalar k: this is true, so they are parallel.

7 [Foundational] • Target: Steady-State (Stationary) Distribution

Simplify $\frac{1-\cos^2\theta}{\sin\theta} \cdot \sec\theta$. .

Final Answer: $\tan\theta$. $1 - \cos^2\theta = \sin^2\theta$, so the fraction is $\sin\theta$. Then $\sin\theta \cdot \sec\theta = \frac{\sin\theta}{\cos\theta} = \tan\theta$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

© ClipSAT — clipsat.org | All Rights Reserved

8 [Intermediate] • Target: Absorbing Markov Chains

Line: $\mathbf{r} = (2; 2; -3) + t(-2; -2; 0)$. Plane: $-4x - 2z = 14$. Find the value of t at the point of intersection.

Final Answer: 2. Working through this vectors step: $\mathbf{n} \cdot (\mathbf{a} + t\mathbf{d}) = 14 \Rightarrow -2 + t(8) = 14 \Rightarrow t = 2$

9 [Intermediate] • Target: Transition Matrices and State Vectors

Line: $\mathbf{r} = (4; -1; 3) + t(3; 2; -3)$. Plane: $-2x - 4y + 3z = 5$. Find the value of t at the point of intersection.

Final Answer: 0. Working through this vectors step: $\mathbf{n} \cdot (\mathbf{a} + t\mathbf{d}) = 5 \Rightarrow 5 + t(-23) = 5 \Rightarrow t = 0$

10 [Intermediate] • Target: Multi-Step Transitions

Line: $\mathbf{r} = (0; 2; 4) + t(2; 0; 1)$. Plane: $-x - 4y + 3z = 7$. Find the value of t at the point of intersection.

Final Answer: 3. Working through this vectors step: $\mathbf{n} \cdot (\mathbf{a} + t\mathbf{d}) = 7 \Rightarrow 4 + t(1) = 7 \Rightarrow t = 3$

11 [Intermediate] • Target: Steady-State (Stationary) Distribution

Find the distance from $P(-1, -13, 23)$ to the plane $-3x - 4y + 12z = -7$.

Final Answer: 26. Working through this vectors step: Distance $= \frac{|-3(-1) - 4(-13) + 12(23) - (-7)|}{13} = \frac{338}{13} = 26$

12 [Intermediate] • Target: Absorbing Markov Chains

Find the distance from $P(-28, -10, 11)$ to the plane $-6x - 2y + 3z = -24$.

Final Answer: 35. Working through this vectors step: Distance $= \frac{|-6(-28) - 2(-10) + 3(11) - (-24)|}{7} = \frac{245}{7} = 35$

13 [Intermediate] • Target: Transition Matrices and State Vectors

Find the distance from $P(-1, 2, -11)$ to the plane $-2x + 6y - 9z = -8$.

Final Answer: 11. Working through this vectors step: Distance $= \frac{|-2(-1) + 6(2) - 9(-11) - (-8)|}{11} = \frac{121}{11} = 11$

14 [Intermediate] • Target: Multi-Step Transitions

Find the distance from $P(-13, -33, 13)$ to the plane $-3x - 6y + 2z = 18$.

Final Answer: 35. Working through this vectors step: Distance $= \frac{|-3(-13) - 6(-33) + 2(13) - (18)|}{7} = \frac{245}{7} = 35$

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Steady-State (Stationary) Distribution • Cross-Lesson: 5.04 Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)

Find the distance from $P(25, 11, 16)$ to the plane $6x + 2y + 3z = -25$.

Final Answer: 35. Working through this vectors step: Distance $= \frac{|6(25) + 2(11) + 3(16) - (-25)|}{7} = \frac{245}{7} = 35$

© ClipSAT — clipsat.org | All Rights Reserved

16 [Challenging] • Target: Absorbing Markov Chains • Cross-Lesson: 6.01 Matrices (AI HL)

Find the distance from $P(-15, -3, -8)$ to the plane $-6x - 2y - 3z = 22$.

Final Answer: 14. Working through this vectors step: Distance $= \frac{|-6(-15) - 2(-3) - 3(-8) - (22)|}{7} = \frac{98}{7} = 14$

17 [Challenging] • Target: Transition Matrices and State Vectors • Cross-Lesson: 6.02 Graph Theory & Networks (AI HL)

Line 1: $\mathbf{r} = (-2; -4; 2) + s(-2; 3; -1)$. Line 2: $\mathbf{r} = (0; -1; 4) + t(-4; 6; -2)$. Are the lines parallel? Show your complete algebraic work.

Final Answer: Yes. Direction vector 2 = 2 × direction vector 1, so the lines are parallel.

18 [Challenging] • Target: Multi-Step Transitions • Cross-Lesson: 1.01 Proof by Mathematical Induction

Line 1: $\mathbf{r} = (0; -1; 3) + s(-2; 4; 1)$. Line 2: $\mathbf{r} = (1; 3; 4) + t(-4; 8; 2)$. Are the lines parallel? Show your complete algebraic work.

Final Answer: Yes. Direction vector 2 = 2 × direction vector 1, so the lines are parallel.

19 [Challenging] • Target: Steady-State (Stationary) Distribution • Cross-Lesson: 1.02 Complex Numbers: Cartesian Form & Algebra

Line 1: $\mathbf{r} = (2; 2; 4) + s(0; 4; -2)$. Line 2: $\mathbf{r} = (-1; -5; 4) + t(0; 12; -6)$. Are the lines parallel? Show your complete algebraic work.

Final Answer: Yes. Direction vector 2 = 3 × direction vector 1, so the lines are parallel.

20 [Challenging] • Target: Absorbing Markov Chains • Cross-Lesson: 1.03 Polar Form, De Moivre's Theorem & Roots of Unity

Line 1: $\mathbf{r} = (-1; 5; -2) + s(2; 5; -3)$. Line 2: $\mathbf{r} = (-1; 4; 0) + t(4; 10; -6)$. Are the lines parallel? Show your complete algebraic work.

Final Answer: Yes. Direction vector 2 = 2 × direction vector 1, so the lines are parallel.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

© ClipSAT — clipsat.org | All Rights Reserved

21 [Foundational] • Target: Transition Matrices and State Vectors

In a standard transition probability matrix T (where entry T_{ij} represents the probability of moving from state i to state j), what must the sum of the entries in each row equal?

a. 1 ✓

b. 0

c. The number of states

d. $\frac{1}{n}$

Because each row corresponds to the probability distribution of transitioning from a given state to all possible next states, the sum of probabilities across each row is strictly 1.

22 [Foundational] • Target: Transition Matrices and State Vectors

Given initial state probability vector $\mathbf{s}_0 = [0.6, 0.4]$ and transition matrix $T = [[0.8, 0.2], [0.3, 0.7]]$, what is the state distribution after one step $\mathbf{s}_1 = \mathbf{s}_0 T$?

a. $[0.60, 0.40]$ ✓

b. $[0.55, 0.45]$

c. $[0.66, 0.34]$

d. $[0.48, 0.28]$

$\mathbf{s}_1 = [0.6(0.8) + 0.4(0.3), 0.6(0.2) + 0.4(0.7)] = [0.48 + 0.12, 0.12 + 0.28] = [0.60, 0.40]$.

23 [Foundational] • Target: Transition Matrices and State Vectors

What is the Markov property (memoryless property) in stochastic processes?

a. **The future state depends only upon the current state, not on the sequence of states that preceded it** ✓

b. The process has zero variance

c. All states have equal probability of occurrence

d. The chain can never visit the same state twice

A first-order Markov chain satisfies $P(X_{n+1} = s \mid X_n, X_{n-1}, \ldots, X_0) = P(X_{n+1} = s \mid X_n)$, meaning the conditional probability distribution depends only on the present state.

© ClipSAT — clipsat.org | All Rights Reserved

24 [Foundational] • Target: Multi-Step Transitions

How is the probability distribution $\mathbf{s}_n$ after n transitions computed from the initial state vector $\mathbf{s}_0$ and transition matrix T?

a. $\mathbf{s}_n = \mathbf{s}_0 T^n$ ✓

b. $\mathbf{s}_n = \mathbf{s}_0 + nT$

c. $\mathbf{s}_n = T^n \mathbf{s}_0$

d. $\mathbf{s}_n = (\mathbf{s}_0 T)^n$

Iterating the transition step $\mathbf{s}_1 = \mathbf{s}_0 T$, $\mathbf{s}_2 = \mathbf{s}_1 T = \mathbf{s}_0 T^2$ leads to $\mathbf{s}_n = \mathbf{s}_0 T^n$.

25 [Foundational] • Target: Multi-Step Transitions

If $T = [[0.5, 0.5], [0.5, 0.5]]$, evaluate T^2.

a. $[[0.5, 0.5], [0.5, 0.5]]$ ✓

b. $[[0.25, 0.25], [0.25, 0.25]]$

c. $[[1, 0], [0, 1]]$

d. $[[0.5, 0], [0, 0.5]]$

Each entry is $0.5(0.5) + 0.5(0.5) = 0.25 + 0.25 = 0.5$. Thus $T^2 = T$.

26 [Foundational] • Target: Multi-Step Transitions

What does entry $(T^3)_{2,1}$ represent?

a. **The probability of transitioning from State 2 to State 1 in exactly 3 steps** ✓

b. The probability of transitioning from State 1 to State 2 in 3 steps

c. Three times the one-step probability

d. The cube of the probability from State 2 to State 1

In the n-step transition matrix T^n, the entry in row i, column j is the conditional probability of being in state j after n steps given starting in state i.

© ClipSAT — clipsat.org | All Rights Reserved

27 [Foundational] • Target: Steady-State (Stationary) Distribution

What defining matrix equation must the steady-state probability vector $\boldsymbol{\pi} = [\pi_1, \pi_2]$ satisfy?

a. $\boldsymbol{\pi}T = \boldsymbol{\pi}$ **and** $\sum \pi_i = 1$ ✓

b. $T\boldsymbol{\pi} = \mathbf{0}$

c. $\boldsymbol{\pi}T = \mathbf{0}$

d. $\boldsymbol{\pi} = T^{-1}$

A steady-state distribution is stationary under transitions ($\boldsymbol{\pi}T = \boldsymbol{\pi}$) and its components form a valid probability distribution (sum to 1).

28 [Foundational] • Target: Steady-State (Stationary) Distribution

Find the stationary distribution for $T = [[0.7, 0.3], [0.2, 0.8]]$.

a. $\boldsymbol{\pi} = [0.4, 0.6]$ ✓

b. $\boldsymbol{\pi} = [0.5, 0.5]$

c. $\boldsymbol{\pi} = [0.3, 0.7]$

d. $\boldsymbol{\pi} = [0.6, 0.4]$

Set $[\pi_1, \pi_2][[0.7, 0.3], [0.2, 0.8]] = [\pi_1, \pi_2] \Rightarrow 0.7\pi_1 + 0.2\pi_2 = \pi_1 \Rightarrow 0.2\pi_2 = 0.3\pi_1 \Rightarrow 3\pi_1 = 2\pi_2$. Using $\pi_1 + \pi_2 = 1$: $\pi_1 + 1.5\pi_1 = 1 \Rightarrow 2.5\pi_1 = 1 \Rightarrow \pi_1 = 0.4$, $\pi_2 = 0.6$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Steady-State (Stationary) Distribution

When is a Markov chain guaranteed to have a unique stationary distribution that is independent of the initial state?

a. **When the transition matrix is regular (some power T^k has strictly positive entries)** ✓

b. Only if the matrix is symmetric

c. Whenever $\det(T) = 0$

d. Only if there are 2 states

By the Perron-Frobenius theorem, a regular Markov chain has a unique stationary distribution, and $\lim_{n\to\infty} \mathbf{s}_0 T^n = \boldsymbol{\pi}$ for any initial distribution $\mathbf{s}_0$.

© ClipSAT — clipsat.org | All Rights Reserved

30 [Intermediate] • Target: Absorbing Markov Chains

What is an absorbing state in a Markov chain?

a. A state that, once entered, cannot be left ($T_{ii} = 1$ and $T_{ij} = 0$ for $j \neq i$) ✓

b. A state that has probability zero of being visited

c. A state that transitions with equal probability to all other states

d. A state where the steady-state probability is 0

An absorbing state has a loop transition probability of 1 to itself, trapping the process indefinitely once entered.

31 [Intermediate] • Target: Absorbing Markov Chains

In an absorbing Markov chain in canonical form $P = [[I, 0], [R, Q]]$, what does the fundamental matrix $N = (I - Q)^{-1}$ calculate?

a. The expected number of visits to each transient state before absorption ✓

b. The probability of absorption in one step

c. The eigenvalues of the transition matrix

d. The total number of absorbing states

The fundamental matrix $N = (I - Q)^{-1}$ provides the expected time (number of steps) spent in each transient state starting from any transient state before reaching an absorbing state.

32 [Intermediate] • Target: Absorbing Markov Chains

How is the absorption probability matrix B calculated from fundamental matrix N and transient-to-absorbing matrix R?

a. $B = NR$ ✓

b. $B = RN$

c. $B = N + R$

d. $B = (I - Q)R$

The matrix of absorption probabilities from each transient state into each absorbing state is given by the matrix product $B = NR$.

© ClipSAT — clipsat.org | All Rights Reserved

33 [Intermediate] • Target: Transition Matrices and State Vectors

Line 1: $\mathbf{r} = (-3; 4; -4) + s(3; -5; -4)$. Line 2: $\mathbf{r} = (2; 4; 1) + t(6; -10; -8)$. Are the lines parallel?

a. They are the same line

b. Yes ✓

c. They intersect at one point

d. No

Direction vector 2 = 2 × direction vector 1, so the lines are parallel.

34 [Intermediate] • Target: Multi-Step Transitions

Find the distance from $P(8, -5, 5)$ to the plane $6x - 2y + 3z = 24$.

a. 5

b. 4

c. **7** ✓

d. 2

Working through this vectors step: Distance $= \frac{|6(8) - 2(-5) + 3(5) - (24)|}{7} = \frac{49}{7} = 7$

35 [Intermediate] • Target: Steady-State (Stationary) Distribution

Find the distance from the origin to the plane $3x + 4y - 12z = 52$.

a. 5

b. 6

c. **4** ✓

d. 2

Working through this vectors step: Distance $= \frac{|d|}{|\mathbf{n}|} = \frac{52}{13} = 4$

36 [Intermediate] • Target: Absorbing Markov Chains

Find the distance from the origin to the plane $3x - 4y + 12z = 52$.

a. 1

b. 2

c. **4** ✓

d. 5

Working through this vectors step: Distance $= \frac{|d|}{|\mathbf{n}|} = \frac{52}{13} = 4$

© ClipSAT — clipsat.org | All Rights Reserved

37 [Intermediate] • Target: Transition Matrices and State Vectors

Find the distance from the origin to the plane $-3x - 4y + 12z = 39$.

a. **3** ✓

b. 4

c. 1

d. 6

Working through this vectors step: $\text{Distance} = \frac{|d|}{|\mathbf{n}|} = \frac{39}{13} = 3$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multi-Step Transitions • Cross-Lesson: 1.01 Proof by Mathematical Induction

Find the distance from the origin to the plane $-3x - 4y - 12z = 13$.

a. 3

b. 2

c. 4

d. **1** ✓

Working through this vectors step: $\text{Distance} = \frac{|d|}{|\mathbf{n}|} = \frac{13}{13} = 1$

39 [Challenging] • Target: Steady-State (Stationary) Distribution • Cross-Lesson: 1.02 Complex Numbers: Cartesian Form & Algebra

Find $\mathbf{u} \cdot \mathbf{v}$ if $\mathbf{u} = (-1; 5; -3)$ and $\mathbf{v} = (-5; -3; -6)$.

a. 4

b. 2

c. 11

d. **8** ✓

Working through this vectors step: $\mathbf{u} \cdot \mathbf{v} = (-1)(-5) + (5)(-3) + (-3)(-6) = 8$

40 [Challenging] • Target: Absorbing Markov Chains • Cross-Lesson: 1.03 Polar Form, De Moivre's Theorem & Roots of Unity

Find $\mathbf{u} \cdot \mathbf{v}$ if $\mathbf{u} = (3; -4; 6)$ and $\mathbf{v} = (2; -2; -6)$.

a. **-22** ✓

b. -23

c. -30

d. -33

Working through this vectors step: $\mathbf{u} \cdot \mathbf{v} = (3)(2) + (-4)(-2) + (6)(-6) = -22$

© ClipSAT — clipsat.org | All Rights Reserved

41 [Challenging] • Target: Transition Matrices and State Vectors • Cross-Lesson: 1.04 Partial Fractions

Find $\mathbf{u} \cdot \mathbf{v}$ if $\mathbf{u} = (-6; 4; -1)$ and $\mathbf{v} = (-6; 4; -1)$.

a. 42

b. **53** ✓

c. 56

d. 50

Working through this vectors step: $\mathbf{u} \cdot \mathbf{v} = (-6)(-6) + (4)(4) + (-1)(-1) = 53$

42 [Challenging] • Target: Multi-Step Transitions • Cross-Lesson: 1.05 Systems of Linear Equations (AA HL)

Find $\mathbf{u} \cdot \mathbf{v}$ if $\mathbf{u} = (6; 6; 5)$ and $\mathbf{v} = (-6; -2; -2)$.

a. -54

b. -69

c. -51

d. **-58** ✓

Working through this vectors step: $\mathbf{u} \cdot \mathbf{v} = (6)(-6) + (6)(-2) + (5)(-2) = -58$

43 [Challenging] • Target: Steady-State (Stationary) Distribution • Cross-Lesson: 2.01 Modulus Functions & the Factor/Remainder Theorem

Find the distance from the origin to the plane $3x + 6y + 2z = 28$.

a. 7

b. 3

c. 2

d. **4** ✓

Working through this vectors step: $\text{Distance} = \frac{|d|}{|\mathbf{n}|} = \frac{28}{7} = 4$

44 [Challenging] • Target: Absorbing Markov Chains • Cross-Lesson: 3.01 3D Vectors & the Cross Product

Find the distance from the origin to the plane $4x - 4y + 7z = 36$.

a. **4** ✓

b. 1

c. 7

d. 2

Working through this vectors step: $\text{Distance} = \frac{|d|}{|\mathbf{n}|} = \frac{36}{9} = 4$

© ClipSAT — clipsat.org | All Rights Reserved

45 [Challenging] • Target: Transition Matrices and State Vectors • Cross-Lesson: 3.02 Lines and Planes in Three Dimensions

Find the distance from the origin to the plane $-4x - 4y + 7z = 18$.

a. **2** ✓

b. 5

c. 4

d. 1

Working through this vectors step: $\text{Distance} = \frac{|d|}{|\mathbf{n}|} = \frac{18}{9} = 2$

© ClipSAT — clipsat.org | All Rights Reserved